

Training Configuration and Summary:

Sequencing:

- timesteps: 30
- batch_size: 128

Model Hyperparameters:

- min_units: 128
- max_units: 256
- units_step: 64
- dropout_min: 0.2
- dropout_max: 0.3
- dropout_step: 0.1
- activation_functions: ['tanh', 'relu', 'elu']
- optimizers: ['adam', 'rmsprop', 'Nadam']

Tuning:

- max_epochs: 30
- hyperband_iterations: 3

LSTM Multi-Step Training Report

Report Generated: 2025-04-14 23:24:52

Training Start: 2025-04-14 19:48:56

Training End: 2025-04-14 20:27:44

Training Duration: 38.79 minutes

Best Hyperparameters:

Number of Layers: 2

Activation: relu

Optimizer: Nadam

Dropout Rate: 0.20

MAE: 1.4255

MSE: 3.9646

RMSE: 1.9911

R2 Score: 0.9100

Multi-Step Forecast:

Forecast Steps: 5

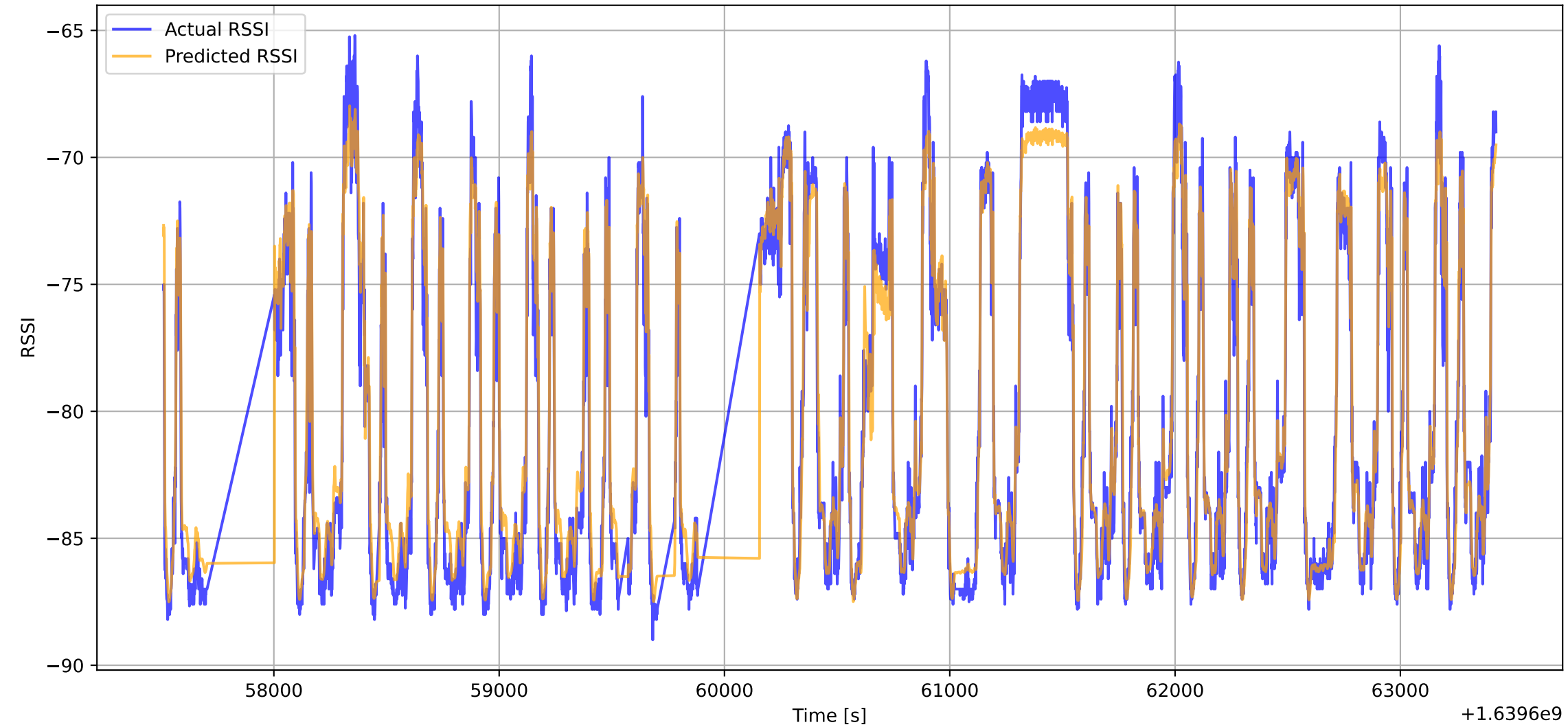
MAE: 0.0310

MSE: 0.0017

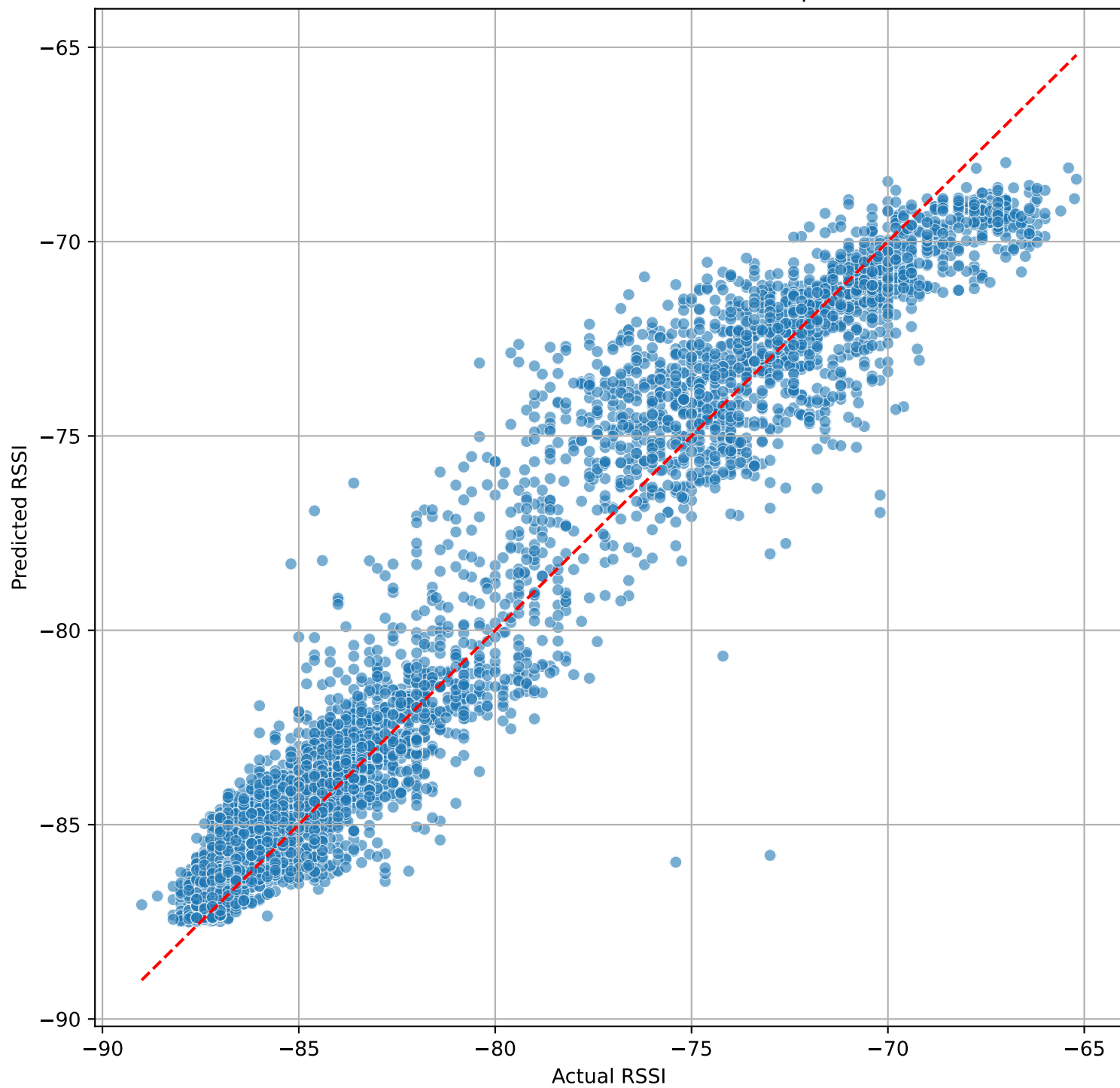
RMSE: 0.0415

R2 Score: 0.9498

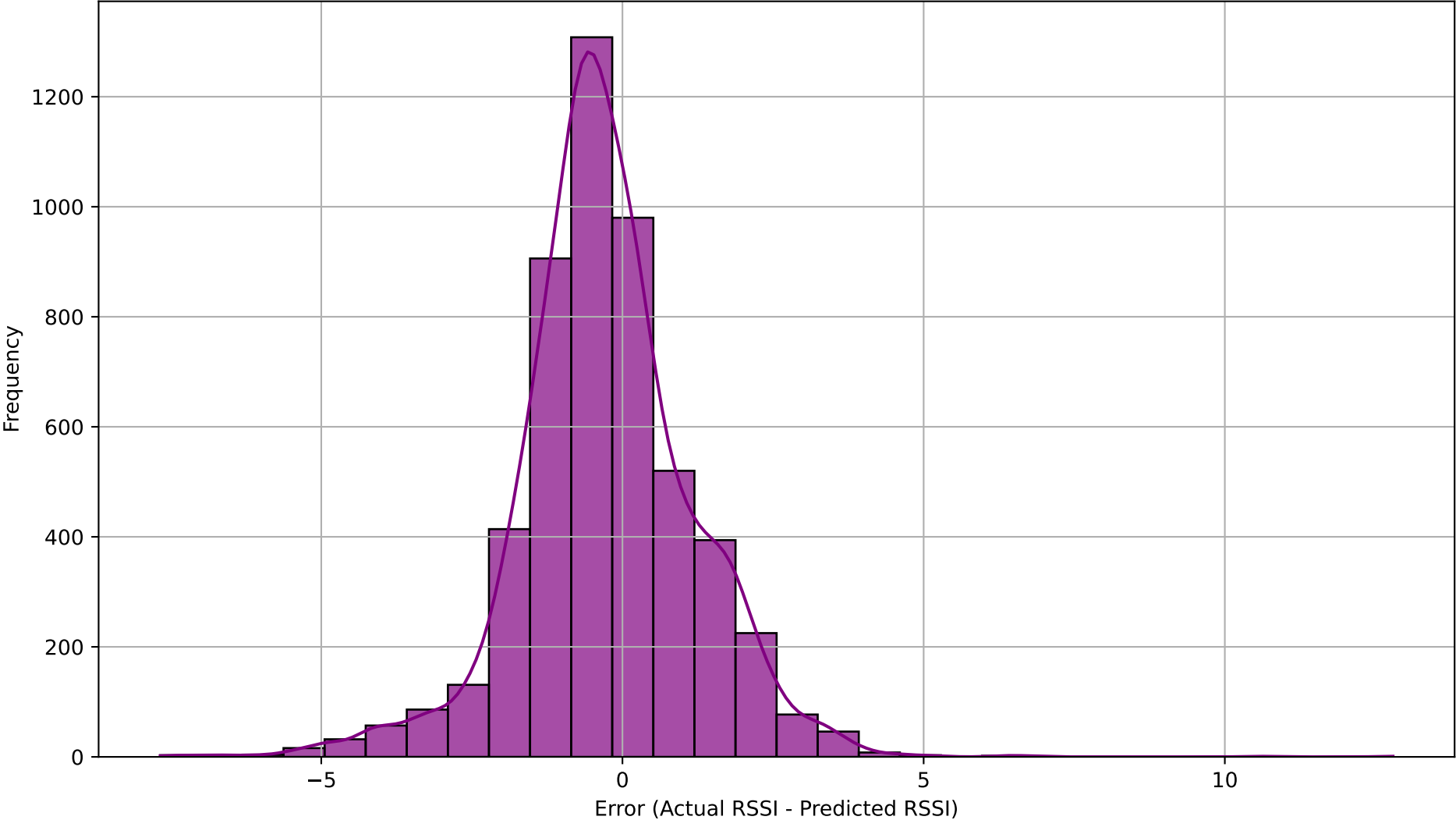
Actual vs Predicted RSSI (First Step)



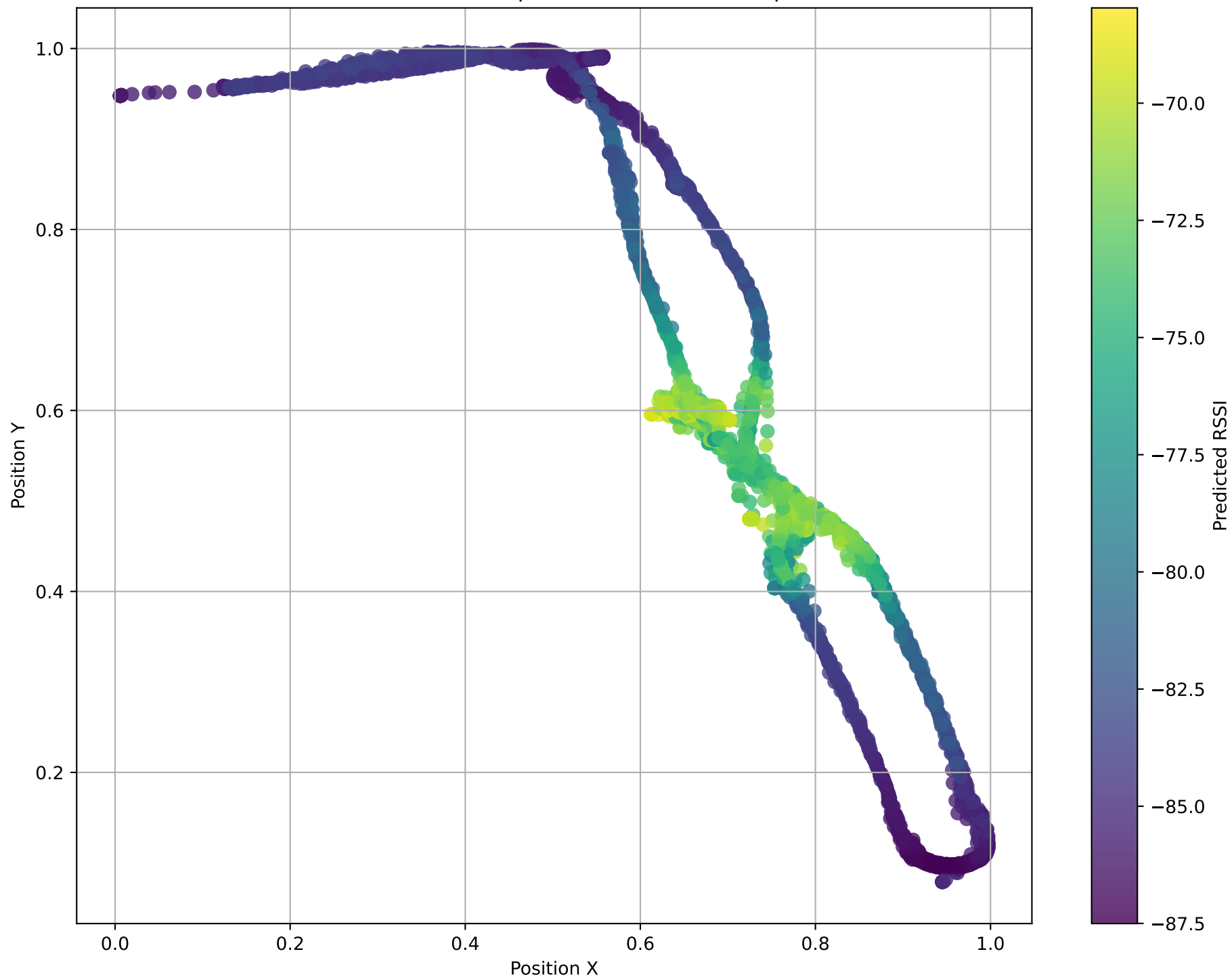
Actual vs Predicted RSSI (First Step)



Distribution of Prediction Errors (First Step)



Predicted RSSI in Spatial Context (First Step)



Multi-Step Forecast Error by Step

